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Conceptual wanderlust: How to develop creative supply chain theory with analogies

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Abstract

This article guides the development of creative insights in supply chain management. The authors begin by describing analogies' role in advancing organization theory with a focus on image transfers between knowledge domains. While much has been said about the merits of conceptual transfers between domains, much equally remains unknown, particularly about how to deliberately develop creative insights that have the potential to challenge or enhance existing supply chain management theory. To address this issue, this research builds on prior work on analogies, creativity, and design thinking to develop a heuristic-analytic model. This model is designed to assist researchers in theorizing by analogy through a controlled dual cognitive process of divergence before convergence combined with a distant boundary-spanning knowledge search. An illustration of the model shows how creative output can provide a fresh perspective that helps render supply chains potentially more resilient.

KEYWORDS

analogy, creativity, supply chain management, theory development

INTRODUCTION

Research on analogies flourishes across the social and related sciences (Gentner & Holyoak, 1997; Hesse, 1966; Jermier & Forbes, 2016). Organization theory is no exception, with much activity drawing on foreign knowledge domains to help simplify abstract concepts, formulate explanations, crystallize thinking, and, importantly, challenge prevailing ways of viewing phenomena (Chen et al., 2013; Cornelissen et al., 2005; Holyoak & Thagard, 1996; Morgan, 2016; Weick, 1989). For example, when scholars view collaborating organizations as a flock of birds (Reeves et al., 2020) or the role of organizational decision-making as akin to that of Alice's Adventures in Wonderland (McCabe, 2016), each analogy conveys a different perspective and a unique way of seeing organizational realities. A similar use of analogies in the early stage of theory development can help scholars indulge in a form of conceptual *wanderlust*, providing intellectual space for exploring foreign knowledge terrains and transferring images from seemingly unrelated domains to solve problems in a familiar target domain (Amabile, 1998; Breslin & Gatrell, 2023; Nadkarni et al., 2018). However, particularly in the supply chain management (SCM) field, using analogies to deliberately generate new and relevant—that is, *creative*—insights that challenge or enhance existing supply chain theory remains limited (Stephens et al., 2022).

Perhaps the relative dearth of creative theorizing via analogies in SCM is not surprising. Creative insights

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stemming from analogies are inherently incomplete and thus open to interpretation. This subjectivity can contribute to a sense of unease, particularly for those scholars taught to be experts at assessing and testing theories to move toward a positivist theoretical consensus (Carter, 2011; Hardy et al., 2020). Doing so can come at the expense of undervaluing the creative potential of the theory discovery process. While critical, theory assessment usually follows the discovery stage of theory development (Darden, 1991). Assessment alone does not often lead to creative insights. For example, Wieland (2021: 58), in his efforts to replace traditional views of SCM with a more contemporary vision of "dancing the supply chain," leverages the creative potential of discovery. It is unlikely that such an insight would emerge from a study designed to primarily assess the validity of a theory. Thus, our proposed process is designed to maximize creativity in the theory initiation (discovery) stage. We do not aim to devalue the importance of theory validation or ignore how a rigorous assessment and testing phase can provide confidence in empirical adequacy. Our model is designed for theory development, not for testing theory. We argue that neglecting the discovery stage that initiates knowledge development leaves the SCM discipline potentially exposed to a form of circular research focused on testing variations of the same theories (Yadav, 2010). In contrast, developing new theory could help SCM scholars and managers to predict, understand, and deal more effectively with emerging issues and potential disruptions, including those related to pandemics, political riots, and intensifying natural disasters (Choi et al., 2020; Schmenner & Swink, 1998; Sinfield et al., 2014).

While some guidance for reasoning by analogy in SCM and adjacent disciplines exists (Chen et al., 2013; Foropon & McLachlin, 2013; Gruner & Power, 2020; Stephens et al., 2022), much remains unknown about how to combine knowledge by transferring images (that is: concepts, laws, theories, or any other element pertaining to a source domain) with creative intent. Thus, in this article, we ask: How can we transfer images between knowledge domains to develop creative insights that challenge or enhance existing SCM theory?

In answering this question, we guide SCM scholars who, in the early stage of theorizing and thought experimentation, seek to use analogies to conceive new theory, not unlike Stephens et al. (2022) and Wieland (2021). Our model aims to facilitate the development of new theoretical insights that can address SCM issues by challenging or enhancing existing theory. In this context, analogies' function is heuristic. Their use assists scholars in introducing new theories with the potential to help deal with both extant and emerging issues. When we refer to "theories" or "theoretical insights," we mean propositional arguments that involve using one concept to explain another (Whetten et al., 2009). More specifically, we develop a heuristic-analytic model that helps scholars deliberately and iteratively transfer images between knowledge domains for creative theoretical insights.

The model has two key defining features: (1) a controlled dual cognitive process and (2) a boundary-spanning knowledge search. The model's first defining feature relates to the importance of a knowledge domain transfer that consists of an intuitive (or "divergent") boundary-spanning knowledge search to generate many early insights (the heuristic part of the model). This knowledge transfer is followed by more effortful (or "convergent") thinking to assess the insights' creative value (the analytic part of the model). The model's second defining feature relates to the importance of using a boundary-spanning knowledge search when engaging in divergent thinking, stretching existing theoretical boundaries in search of explanations beyond our own discipline by drawing on conceptually dissimilar knowledge domains to generate creative insights (Gulati, 2007; Nadkarni et al., 2018; Stephens et al., 2022). We illustrate our model by drawing on conceptually distant source phenomena pertaining to the natural sciences. Specifically, through phenomena in physics (e.g., entropy) and biology (e.g., critical transitions), we showcase an alternative approach to supply chain resilience and the management of disruptive events (Altay & Ramirez, 2010; Brandon-Jones et al., 2014).

Among other contributions, we seek to equip SCM scholars with knowledge on developing creative theory (Darden, 1991; Yadav, 2010). We thus add to a growing body of work concerned with different ways to develop supply chain-specific theory (Fugate et al., 2019; Hardy et al., 2020; Matthews et al., 2016). Further, we build upon and contribute to a small body of work focused on analogical reasoning guidance in SCM (e.g., Chen et al., 2013; Gruner & Power, 2020). Therefore, we also aid scholars—somewhat ironically—in developing their own theoretical foundations by engaging in a form of conceptual *wanderlust*, that is, explorations of foreign knowledge terrains (Nadkarni et al., 2018). Finally, we illustrate our model with resilient SCM acting as theorizing beneficiary (i.e., target domain). We thus respond to calls for knowledge advances that can help managers navigate disruptive events, such as those related to COVID-19 (Craighead et al., 2020).

BACKGROUND

We speak of analogies whenever we think of a phenomenon as something else. Analogies allow us to create a

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bridge between two knowledge domains: the source and the target domain (Gentner & Markman, 1997). Analogies also play a vital role in the theorizing process, as they help us comprehend and interpret the world around us (Cornelissen & Kafouros, 2008; Stephens et al., 2022). When we theorize by analogy, we use the source domain to learn about a target domain and view various phenomena by presenting them "as if they were X" (Cornelissen, 2004; Morgan, 1980; Morgan, 1983b; Morgan, 1986; Tsoukas, 1991). The focus here is usually on similarities, as an analogy makes us attend to some likeness (ideally, a surprising and somewhat counterintuitive likeness between two or more concepts). At the same time, analogies permit us to ignore irrelevant similarities and dissimilarities (Oswick et al., 2002).

More to the point, analogies can offer new ways of seeing and thinking as they enhance our creative abilities, helping us to better intuit and deal with the many facets of organizational life (Morgan, 1986). As such, analogies can serve as a provisional way of organizing and understanding complex phenomena, providing a precursor to more formal theorizing and observations (Cornelissen, 2005). For example, drawing an analogy between a business and a living organism can offer fresh perspectives on the interrelationships between trading partners, their operations, and their ability to adjust to changing environments. As theory and research advance, the living organism perspective could become so ingrained that the original imagery is overlooked, leaving behind residual concepts, such as the notion of "ecosystems" or "network resilience" that continues to be the subject of extended theorizing and research.

Many of the most influential organizational and management theories are based on a core analogy (Ketokivi et al., 2017). For example, the resource-based view of the firm proposes that firms' resources and capabilities are key determinants of their competitive advantage (Penrose, 1995). The foundational analogy here is that how a person's skills, knowledge, and abilities impact their success is akin to the way organizations' resources and capabilities determine their performance in a given market.

Not surprisingly, then, management researchers' interest in analogies takes many forms, often reflecting broader meta-theoretical issues around theorizing, research, and knowledge development (Bacharach, 1989; Cornelissen et al., 2005; Suddaby et al., 2011; Weick, 1989). Deliberately using analogies in the early stage of knowledge development to challenge and enhance organization theory is not new either and falls within the process-based view of theorizing (Rindova, 2011). Transferring information from a source onto a target domain is an imaginative process that can guide us toward new questions and insights, and thus potentially also creative theory advances (Alajoutsijärvi et al., 2001; Breslin & Gatrell, 2023; Cornelissen, 2005; Rindova, 2011; Schön, 1979). The following sub-section further explores the link between creativity, theorizing, and knowledge domain transfers.

Creativity, theorizing, and knowledge domain transfers

Scholars' capacity to build upon existing knowledge to make a theoretical contribution relies on their ability to produce original and useful (in short: *creative*) insights. The greater the degree of novelty, revelation, and surprise in their insights, the more likely scholars are to meaningfully contribute to their field (Corley & Gioia, 2011). In theorizing by analogy, such creative insights are tied to transferring images from a source domain to rearrange, advance, or challenge our understanding of a target phenomenon (Cornelissen, 2005). For example, scholars might ask whether organizational rumors spread like diseases and thus study the principles of epidemiology as a model of rumor transmission (Smith, 1981). Other examples include studying the human brain to understand better how organizations might design flexible production systems (Garud & Kotha, 1994) or whether root-fungus networks might hold insights on transitioning operations from potentially wasteful uni-directional and linear processes to more sustainable, circular ones (Tate et al., 2019). However, much remains to be learned about how to transfer images between domains for creative output that can exert a theory-formative influence in the initiating, early theory development stage.

With this in mind, we propose a heuristic-analytic model that introduces a sequence of divergent and convergent phases (or iterations), each concerned with combining knowledge domains for creative insights. After identifying a target domain and associated research problem(s), the model triggers divergent thinking in each iteration to learn from conceptually distant domains through effortless or heuristic (i.e., divergent) thinking. This phase is followed by convergent thinking to assess the ideas' creative value and translate (that is, tie down and articulate) the most promising ideas into creative, theoretical insights. Together, the two phases make up an iteration where the output of each phase becomes the input for the next. These two phases are re-iterated as many times as necessary. The following section introduces each defining feature separately before outlining how they underpin our model.

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ANALOGIES AND CREATIVE THEORIES

Controlled dual cognitive processes

Consistent with dual processing theories and the design thinking method, creativity is associated with both divergent and convergent thinking (Auernhammer & Roth, 2021; Beckman & Barry, 2007; Osborne, 1957). These two cognitive modes are partners in the construction of meaning (Evans, 2008) and have been labeled in many different ways, including ideation versus evaluation (Basadur, 1995) and fast/intuitive versus slow/rational thinking (Kahneman, 2011). While divergent thinking has received more research attention, both types are acknowledged as playing a pivotal role in generating insights.

Controlling whether to engage in one or the other process is critical in generating creative output through image transfers between knowledge domains. When encountering unfamiliar source domains, most impressions and ideas will probably be blocked or corrected without first restraining the more critical convergent thinking mode (Morewedge & Kahneman, 2010). While this gatekeeping is often necessary to avoid being overwhelmed by combinatorial choices and possibilities, it should initially be kept at bay as our capability to generate *and* judge insights concurrently is limited (Silvia, 2008).

However, scholars can enhance their creativity while engaging in divergent thinking if they temporarily restrain their critical, convergent thinking mode. Through this process, individuals can explore new perspectives and generate a wide range of early and creative insights. When entertaining the potentially boundless insights pertaining to as many as hundreds of different source domains (in the divergent thinking mode), it is necessary to rely on a combination of heuristics, experience, and intuitions that expedite cognitive processes. But once insights are beginning to crystalize, they should be subjected to more effortful, evaluative convergent thinking with the research problem in the target domain in mind. Many early insights can then be assessed for novelty and relevance.

Boundary-spanning knowledge search

As noted, when theorizing by analogy for creative insights, we also propose that there are benefits of searching for distant (or "boundary-spanning") knowledge that is conceptually dissimilar to one's own (Colquitt & Zapata-Phelan, 2007; Nadkarni et al., 2018; Oswick et al., 2011). While several studies assert the importance of optimal overlap by selecting conceptually similar *and* different source and target domains (e.g., Cornelissen, 2005; Gruner & Power, 2020), this approach is often not conducive to generative, creative impact in the early knowledge development stage (Oswick et al., 2002). In contrast, interdisciplinary think tanks (such as IDEO and the Santa Fe Institute) demonstrate that combining knowledge from different fields has the potential to generate valuable synergies and creative ideas (Fleming, 2004). Breaking free of theorizing that is too firmly based on existing knowledge is also one of the foundational principles of the abductive path of reasoning (Gioia et al., 2012). This approach involves observing diverse phenomena rather than relying solely on current theories, which can potentially result in new insights, explanations, and, ultimately, new theory (Alvesson & Kärreman, 2007).

Consider, for example, if one were to study circular supply chains as particular kinds of spider webs. Doing so could, at first blush, seem too much of a conceptual contortion. Researchers might then easily succumb to an instinctive reaction of "this can't possibly work" when turning toward such a distant domain (Glen et al., 2014). However, ruling out spider webs as too foreign and unrelated to circular supply chains could forfeit insights and parallels stemming from this counterintuitive blend of knowledge domains that integrate ideas of sustainability, freedom, prey, and control. Especially when working with distant domains, many differences need to be, ultimately, ignored. Unless, of course, differences prove to be fruitful. For example, could it be beneficial for some supply chains, or parts thereof, to renew like a spider web, or should they always be built to last? In the next section, we lay out our heuristic-analytic model to provide guardrails for anyone seeking to step outside habitual ways of theorizing by combining knowledge domains for creative output.

COMBINING KNOWLEDGE DOMAINS FOR CREATIVE INSIGHTS

Three simple questions help guide the analogical theorizing process we put forth: (1) "What is?" is concerned with identifying a target domain; (2) "What if?" is concerned with identifying a wide range of possible ideas from distant domains in a divergent manner to benefit the target domain; and (3) "What works?" is concerned with selecting and making sense of the most promising ideas in a convergent manner. Iterating between divergent and convergent modes of thought to drive innovation and

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creative problem-solving is not new (e.g., Runco & Acar, 2012). The two thinking modes play a crucial role in the design thinking method, particularly the Double Design innovation process popularized by the UK Design Council (2015). In building and explicating the steps in our model, we lean on this approach and qualitative research guidance in the organizational literature (e.g., Brown, 2008; Glen et al., 2014; Grodal et al., 2021).

Step 1: identifying a target domain ("What Is?")

The first step in image transfers between knowledge domains is the identification of a target domain, which usually coincides with the researchers' focal knowledge area(s). Once a target domain has been chosen, an associated research question or multiple interrelated research questions should be identified and justified. In so doing, phenomena and issues within a broad knowledge domain often emerge. Indeed, a clearly defined research problem and question are critical in providing a boundary and purpose for the analogical transfer process and are consistent with guidelines for conducting qualitative research (Glaser & Strauss, 1967; Strauss, 1987).

We suggest scholars focus on relevant, emergent, and underexplored real-world issues to tackle significant unresolved problems and refute existing assumptions. One way to do so is by observing conflict between an established body of literature and the real-world, and asking if and how theorizing might lead to actionable advice (Golder et al., 2023). As per our model, identifying problems and asking questions relevant to the academic and practitioner community is necessary to initiate fruitful correspondence between knowledge domains. It might also prove helpful to break down chosen elements in the target domain to ensure a proper understanding of its phenomena and issues, possibly at multiple levels of analysis (Morgeson & Hofmann, 1999). For example, if the target domain is inter-firm relationships, this could be broken down into relevant components of interest, such as strategic alliances, supplier integration, collaboration, trust, transaction costs, and so forth. Doing so could provide a sharper focus in the subsequent stage and jump-start the knowledge transfer process by directing mapping efforts toward a specific theoretical end.

In summary, we propose that scholars ask the question "What is?" to identify a target domain, relevant and specific research problem(s), and constituent elements at multiple levels of analysis; this question is aimed at examining and understanding the status quo, problems, and knowledge (including gaps) that not only give the theorizing a purpose but also serve as a reality check when transferring insights from the source onto the target domain. Doing so allows for an exploration of conceptual possibilities in the divergent and convergent theorizing steps, increasing the chances of putting forward creative theoretical insights that challenge or enhance existing SCM knowledge.

Step 2: the divergent phase ("What If?")

Next, theoreticians should seek potential answers to questions and solutions to research problem(s). As noted, although some solutions could be readily accessible to experts in a field (Nelson & Winter, 1982), a more cognitively challenging, boundary-spanning knowledge search is likely to result in more creative extensions of existing knowledge. This effort is the first foray into the divergent thinking mode, inviting the researcher to take an unconstrained view of possible solutions to the identified research problem by holding off on analytic, evaluative, and critical thinking, for now.

Consistent with the design thinking method (e.g., Brenner et al., 2016), the divergent thinking phase can lead to the adoption of a new lens through which to view a subject, which allows for many early insights to emerge that exert a potentially theory-formative role in the target domain. When deliberately engaging in such boundary-spanning search, researchers will probably have to approach a target problem from a different angle and challenge conventions (Aggarwal & Woolley, 2019; Pollok et al., 2021). To envision multiple options to find solutions and advance theory in a target domain, researchers might find it helpful to ask "What if?" when assessing a wide range of possible ideas and early insights stemming from a distant source domain and its various phenomena. A puzzling, perhaps even perplexing, image in a source domain that challenges the way we view the world (Sloman et al., 1998), and thus also supply chain phenomena, might function as a "lightning rod"—like a puzzling piece of data when developing theory through qualitative research (Grodal et al., 2021: 598)—attracting insights that eventually become the building blocks of creative theories.

Step 3: the convergent phase ("What Works?")

Once the "What if?" question has been addressed, early insights should be assessed in a more effortful convergent thinking mode. When theorizing for creative output, this assessment happens mainly against creativity's two main criteria: novelty and relevance. When evaluating

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previously identified possible insights from distant source domains, a helpful question may be "What works?". Assessing what works means identifying the most promising insights and solutions when exploring distant domain knowledge.

Making calls as to which knowledge domain or insight might yield the most creative solution can be uncomfortable as researchers must venture beyond the confines of their focal knowledge areas. Doing so also involves much-needed simplification. Else, source domain knowledge can prove too ambiguous, complex, and broad to be fully understood or represented. Without such simplification, it is possible to face mental overload that might feel somewhat like what Pettigrew (1990: 281), in a similar context, describes as "a slow and inexorable sinking into a swimming pool which started so cool, clear, and inviting and now has become a clinging mass of maple syrup."

Simplification in the convergent phase happens when source domain knowledge is translated into insights and, eventually, creative insights. Initially, this process is often incomplete and tentative. We suggest focusing on those insights that seem at odds with observations and knowledge in a target domain. Qualitative scholars have called such insights unusual but also acknowledged their importance in the creative theorizing process (Grodal et al., 2021).

Iterations of Steps 2 and 3: Divergence–convergence

As outlined, turning toward conceptually distant source domains to foster creative theorizing without a convergent phase will probably lead to many new but potentially useless bits of information with little heuristic value. Figure 1 shows how the divergent and convergent phases, although distinct, are interrelated and how several iterations may be necessary to test and refine insights. With each iteration, the outputs or insights from

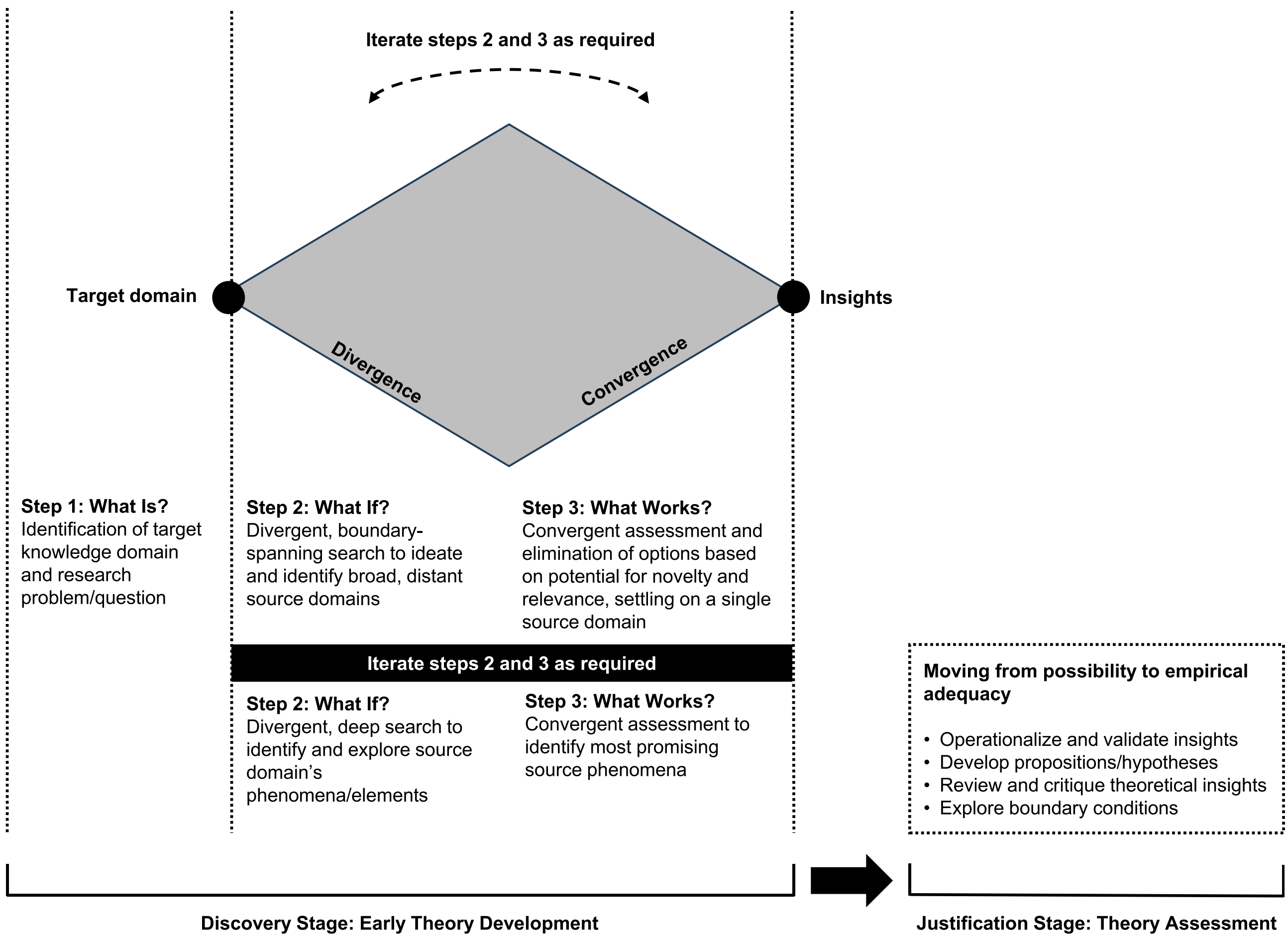

FIGURE 1 The analogical theorizing process for creative output: An approximation.

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the previous iteration become new inputs, progressing scholars through divergence and convergence to increasingly refined and more focused outcomes.

The first step ("What is?") jump-starts the following iterations. The principle behind each iteration is the same: an initial free-flowing, blue-sky phase of divergent thinking helping identify potential ideas from distant source domains. This is followed by a convergent thinking phase to evaluate the options and settle on one or more of the most promising insights. This process is reiterated as often as needed for ever more focused explorations through recurring conceptualization refinements (Gabora, 2010). In this manner, researchers avoid shooting down domains too soon while avoiding an unfocused flight of fancy. At the same time, scholars are also less inclined to simply "borrow" terms and concepts from the source domain and more likely to stay engaged in a form of genuine, two-way exchange between domains (also referred to as a "correspondence approach") (see Oswick et al., 2011).

AN ILLUSTRATION

Next, we illustrate how SCM scholars can transfer images between knowledge domains to develop creative insights in early theory development. We begin by delineating the target domain ("What is?") followed by two divergence–convergence iterations (addressing "What if?" and "What works?").

Identifying a target domain (Step 1)

We have chosen organizational networks as a target domain. Within this domain, we are interested in relations between trading organizations and how to render supply chains more resilient vis-à-vis disruptions. Supply chain disruptions have been identified as one of the most pressing concerns in today's global marketplace (Kocabasoglu-Hillmer et al., 2023; Shekarian & Parast, 2021), so cultivating resilience is a top priority for many organizations. Indeed, supply chains are particularly vulnerable to disruptive challenges, including pandemics, political riots, and intensifying natural disasters (Choi et al., 2020).

Resilience is also a mature theme. Research within a mature and established topic is better suited for illustrating our model and generating new theoretical perspectives (Hardy et al., 2020). Within a mature topic, knowledge advances have the potential to add to and refute previously held assumptions and tackle significant unresolved problems, consistent with our article's intent.

Divergence–convergence (Steps 2–3; Iteration 1)

Once set on a target domain and research problem, a qualitative divergent thinking process was used to choose a distant source domain, operating in an exploratory and investigative fashion consistent with critical tenets of theory development work in the early stage of knowledge creation (Darden, 1991). At its core, selecting a source domain involved new *if-then* propositions aimed at identifying antecedents, consequences, mediators, and moderators that emerged from possible adaptations of knowledge pertaining to potential source domains. Indeed, we entertained various source domains and research fields by consulting the Australian and New Zealand Standard Research Classification. In this early free-flowing divergent thinking stage, we mainly found the inherent tension in transferring information between the two distant domains to be a playful challenge.

In the convergent thinking phase, we dismissed some knowledge domains as conceptually too similar. However, most domains were ruled out because they lacked the potential to yield either original or relevant (i.e., creative) insights. For example, musicology—which can be explored through mathematical relationships and principles associated with harmony—failed to meet the relevance criterion as transferred images seemed more valuable in advancing knowledge on conflict resolution among trading organizations rather than resilience.

Consistent with our model, we moved from divergent to convergent thinking in recurring, ever-more focused conceptualization–refinement iterations (Gabora, 2010; Goldschmidt, 2016). We ultimately settled on natural phenomena pertaining to the fields of physics and biology as source domains relevant to the problem of creating resilient supply chain operations. We found these domains suitable for meeting our key criteria but also theoretically rich in relation to resilient operations in that they could help develop more knowledge about organic models of management processes that are suited to a complex, interconnected, and often disrupted organizational landscape (Farjoun, 2002; Hunt & Menon, 1995). Further, the natural sciences, particularly biology, have previously contributed to knowledge development in organizational disciplines, including supply chains (e.g., Gruner & Power, 2017; Tate et al., 2019).

Divergence–convergence (Steps 2 and 3; Iteration 2)

Once we settled on the source domains, we explored potential new insights through knowledge bridges

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between these unfamiliar domains and our identified research problem. This divergent thinking process was also qualitative in nature. We operated in an exploratory and investigative fashion. We also again put forth *if-then* propositions aimed at identifying antecedents, consequences, mediators, and moderators that emerged from possible adaptations of the source domain knowledge base. However, instead of selecting a broad source domain, in this second iteration, we focused on specific phenomena within our source domains and how these might lead to creative insights that challenge and advance knowledge on resilient supply chains.

In the next phase, we again rejected some of the key principles and phenomena of each source domain while thoroughly examining others. We used initial translations (i.e., mapping of source domain phenomena onto target domain issues) to explain how and where these phenomena may be potentially relevant to our target domain problem, resulting in insights at various levels of generalization (Garud & Kotha, 1994; Morgan, 2016; Tsoukas, 1991). We found that analogical insights were too abstract to be meaningfully evaluated and articulated without such an initial translation process (which requires mental flexibility and adaptation to the source domain). For example, evolutionary theory inspired thought experiments that helped us identify similarities between evolving organisms and supply chains. These include variations in member practices that promote adaptive change and greater resilience over time.

Table 1 features an overview of the divergence–convergence process, including an initial translation and elaboration process for six distant source phenomena that we assessed as most promising (three belonging to physics and three to biology). We first explored the question: What does the principle/phenomenon in the source domain mean ([Q1] in Table 1)? The second question was: What early insights for the target domain (i.e., resilient supply chains) can be gleaned [Q2]? This question invariably led to an initial translation, elaboration, and early insights. The third question was about creativity's novelty element, as we asked: Are insights new to the study of resilient supply chains [Q3]? More specifically, to assess the novelty of emerging insights, we asked if these were original and unexplored with the potential to lead to outcomes that extend or challenge conventional knowledge about resilience in SCM. As such, the novelty criterion relates to whether "potential insights merely add to the momentum created by existing voices, or if they cause heads to turn as the conversation darts in a new direction" (Colquitt & George, 2011: 433). Interestingly, as shown in Table 1, novelty was never categorically present or absent for the most promising source domains. The fourth question was about creativity's relevance element: Can source domain ideas benefit resilient SCM and lead to multiple interesting research questions [Q4]?

MOVING FROM POSSIBILITY TO EMPIRICAL ADEQUACY

Can insights be observed and validated?

Our process model is designed with the intent to challenge or enhance existing knowledge and thus also replace or complement existing SCM theory. Therefore, we intend to refrain from confirming or validating existing SCM theory. In essence, our model's focus is not on testing theory but rather on its development. Our process primarily relies on a subjective image transfer approach, which aligns with the non-positivist research tradition that places less emphasis on validity or generalizability than more objectivist approaches (Bansal & Corley, 2012; Cornelissen et al., 2005; Gentner & Markman, 1997).

Our approach is designed to maximize creativity in the theory initiation stage. Beyond theory initiation, however, a stage of justification or theory assessment is necessary (see Figure 1). This stage is about assessing whether insights are not only creative and plausible but credible and reliable (Darden, 1991; Weick, 1989; Yadav, 2010). At some point, it is essential to consider whether theory (or "second-order concepts") helps predict and shape organizational reality (or "first-order concepts") (Van Maanen, 1979). Put differently, do the creative theoretical insights generated align with empirical observations? This question is crucial because analogies have a dark side in that they can misrepresent the truth, thus obscuring and distorting knowledge development (Oswick et al., 2002). To avoid this, thought experiments in the convergent thinking mode can initially help gauge factual validity and whether theoretical insights can be subjected (at least in principle) to empirical inquiry (Ketokivi et al., 2017). Next, transferred images and often-vague ideas could be tied down and articulated in propositional arguments, formal propositions, or hypotheses (Weick, 1989; Whetten et al., 2009). Doing so means that well-specified, theoretical propositions can be tested qualitatively through field observations and interviews (Gioia et al., 2012; Morgan, 1983a) or by way of quantitative analyses, for example, with a regression model (Ketokivi et al., 2017). At this juncture, criteria typically associated with more positivist research—such as validity and generalizability—can become applicable.

Without empirical validation, scholars are simply operating in the realm of plausibility, with analogies potentially facilitating the introduction of false insights.

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TABLE 1 An illustration of divergence–convergence for image mapping (Steps 2 and 3).

	"What If?"		"What Works?"	
Distant source domains and principles/phenomena	**[Q1] What does the principle/phenomenon in the source domain mean?**	**[Q2] What early insights for the target domain (i.e., resilient supply chains) can be gleaned? (initial translation and elaboration)**	**[Q3] Are insights new to the study of resilient supply chains? (novelty criterion)**	**[Q4] Can source domain ideas benefit resilient SCM and lead to multiple interesting research questions? (relevance criterion)**
Physics				
Entropy	The degree of disorder or randomness in a system (Morowitz, 1955). It represents the unavailability of a system's thermal energy for conversion into mechanical work (Dincer & Cengel, 2001). Lack of order or predictability; gradual decline into disorder (Kumar et al., 2021). The second law of thermodynamics says that entropy always increases with time.	Supply chains operating as systems could become less predictable and more unstable over time. The assumption that resilience is achievable through an equilibrium (or stable state) for a supply chain system could be false and misleading. Equilibrium is usually promoted as a function of successfully implementing "best practice" in supply chain operations. Equilibrium is also traditionally associated with resilient operations (and other desirable "end states" like *sustainable* supply chains). Entropy questions the validity of these states as an achievable objective and implies that ongoing management is required to sustain such states.	*Yes*—The assumption that a desirable and stable state can be defined and brought about through "best practice" in supply chains is not usually questioned. Entropy suggests "better practice" as a more realistic, alternative paradigm. Suppose the notion of "best practice" does not hold. In that case, we must also question the assumption that a genuinely resilient (or even sustainable) stable supply chain system is viable. *No*—The concept of continuous improvement already assumes that whatever is defined as "best practice" at any point is subject to revision over time.	*Yes*—Related research questions include: • Is pursuing stability/equilibrium as implied by the managerialism paradigm (Klikauer, 2015) and "best practice" futile? • Should supply chain managers anticipate disorder and prepare for decline rather than pursue continuous order/truly resilient and enduring systems? • Can we assess the degree to which a supply chain system could decay over time, exposing it to varying degrees of different risks? • What could decay/risk metrics look like (e.g., turnover rate of vendor base; turnover rate of staff at critical supplier firms; supply and demand fluctuations; rate of technological change?) • Can entropy and decline in supply chain systems be mitigated; if so, how?
Resonance	The phenomenon of increased amplitude occurs when the frequency of an applied force is similar to the natural frequency	Supply chain disruptions could result from small changes in phenomena acting on that system. Hence, if it were possible to understand and	*Yes*—The analogical equivalent of resonance in a supply chain context remains relatively unexplored. Predicting, enhancing, avoiding,	*Yes*—Related research questions include: • Which supply chain elements have the potential to trigger significant resonant effects

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TABLE 1 (Continued)

	"What If?"		"What Works?"	
Distant source domains and principles/phenomena	[Q1] What does the principle/phenomenon in the source domain mean?	[Q2] What early insights for the target domain (i.e., resilient supply chains) can be gleaned? (initial translation and elaboration)	[Q3] Are insights new to the study of resilient supply chains? (novelty criterion)	[Q4] Can source domain ideas benefit resilient SCM and lead to multiple interesting research questions? (relevance criterion)
	of systems/materials ("Resonance", 2023). If the system's resonant frequencies are understood, small force applications near this frequency can significantly leverage stored energy to change systems. The integrity of some systems can be compromised by oscillations whose frequencies are close to their natural frequencies as they absorb/store this energy. Dampers can be incorporated into system design to counteract this effect by dissipating this excess energy ("Resonance", 2023).	locate these small changes, it could be possible to intervene and design "dampening" mechanisms (thus enhancing a supply chain's resilience). The resonance principle is related to critical transitions and thresholds (see also below) in that it might be possible to understand how and why small changes in a system lead to significant impacts and even tipping points when supply chains break.	leveraging, "dampening," or recovering from disturbance that led to altered organizational realities offers many novel research avenues. To date, much work on measuring and managing risk across the supply chain, and improving operations, seems based on understanding high-impact strategies and events, not small but critical effects across systems. *No*—the concept of resonance in a supply chain is similar to leverage points used in system dynamics modeling (Sterman, 2002).	following small changes or disruptions? • Can studying minor effects and changes help predict, avoid, or reverse transitions to undesired states? • Is distinguishing between positive and negative resonant effects in a supply chain system possible? • Can supply chain systems be organized in a manner that allows systems to "dampen" negative and maximize positive resonant effects such that systems become more resilient (is there such a thing as a resonant frequency of a supply chain)?
Uncertainty principle	A principle derived from quantum mechanics, whereby a limit exists to the accuracy with which the values for specific pairs of physical quantities of a particle (e.g., its position and momentum) can be predicted from initial conditions (Robertson, 1929). By the same token, the more precise one measurement is, the less precise a complementary measure becomes. Quantum mechanics illustrates the paradoxical behavior of	Supply chains are complex and dynamic, consisting of many interconnected parts. Researchers often build complex models to unravel this complexity and optimize their resilience (among other things). The inherent uncertainty associated with such models, given that what is left out is often critical (analogous to not knowing the effect of such optimization on variables treated as "externalities," i.e., unintended consequences), could call into	*Yes*—a unidirectional design-based logic remains the dominant paradigm for coordinating supply chain activities (Farjoun, 2002). Risk is often seen as measurable, predictable, and manageable at an individual and organizational level rather than at a holistic one that considers diversity and interconnectivity among agents and their role in adapting to changing conditions. If we accept that there is a level of uncertainty inherent in measuring such	*Yes*—Related research questions include: • When faced with disruptions, is it realistic to assume that levels of complexity can be measured accurately so firms can identify and attempt to implement optimal levels of diversity and interconnectivity? • Can the planning/master-design-based logic be reconciled with the emergent paradigm inherent in nature's complex systems when measurement is always problematic, and paradoxes abound?

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TABLE 1 (Continued)

Distant source domains and principles/phenomena	"What If?" [Q1] What does the principle/phenomenon in the source domain mean?	"What If?" [Q2] What early insights for the target domain (i.e., resilient supply chains) can be gleaned? (initial translation and elaboration)	"What Works?" [Q3] Are insights new to the study of resilient supply chains? (novelty criterion)	"What Works?" [Q4] Can source domain ideas benefit resilient SCM and lead to multiple interesting research questions? (relevance criterion)
	physical particles at the sub-atomic level such that nature can be described as absurd when viewed through the lens of common sense (see Feynman, 2006).	question the value of such modeling. If nature is "absurd," then the assumption that measurement can clarify complex supply chains may also be absurd. Many supply chain problems are paradoxical, requiring not so much a measurement approach as questioning key assumptions (e.g., the paradox of free market economies requiring continuous growth vs. resources scarcity in natural systems).	risks, and we also accept that entropy (see above) makes such risks less predictable and difficult to manage, then the value of design-based approaches is questionable. *No*—supply chains have been identified as complex adaptive (and emergent) systems (Tate et al., 2019) where measuring phenomena with single-response surveys is inadequate (Flynn et al., 2018). Also, heuristics are used to create solutions to complex problems as an alternative to optimization models.	• What role might abductive case research play in unraveling (supply chain-related) research paradoxes and accommodating the "absurd" nature of many complex problems? For example, could starting from the paradox itself, such as pursuing economic growth within the limits of natural systems, lead to potential explanations and solutions?
Biology				
Dormancy	A period in natural organisms when growth and development stop to avoid stressful environmental conditions (Baskin & Baskin, 2004). Organisms minimize metabolic activity to conserve energy. The period varies in length and degree of metabolic reduction and can be proactive.	Supply chain members could withstand adverse conditions by making fewer demands on the environment by anticipating and preparing for potential periods of reduced activity. Proactively assessing who or what can be put on hold or reduced and for how long.	*Yes*—Putting parts of its operations into a dormant stage is relatively novel in a supply chain context. It is likely initially perceived as counterintuitive and threatening businesses' ability to react to disturbances. *No*—on the lower end of the dormancy spectrum, supply chains have long followed some of the basic principles (e.g., reducing production capacity in response to often unexpected lower customer	*Yes*—Related research questions include: • Can trading organizations put their operations (or parts thereof) on hold to enhance their resilience when faced with disturbances? • As dormancy types differ in nature and throughout the seasons (e.g., hibernation, diapause, aestivation, and brumation), can these types be meaningfully mapped in an organizational context and

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TABLE 1 (Continued)

	"What If?"		"What Works?"	
Distant source domains and principles/phenomena	[Q1] What does the principle/phenomenon in the source domain mean?	[Q2] What early insights for the target domain (i.e., resilient supply chains) can be gleaned? (initial translation and elaboration)	[Q3] Are insights new to the study of resilient supply chains? (novelty criterion)	[Q4] Can source domain ideas benefit resilient SCM and lead to multiple interesting research questions? (relevance criterion)
			demand) (Chen & Paulraj, 2004). What is novel, then, is considering more extreme forms of dormancy.	seasonal business cycles and activity? • How are dormancy, self-sufficiency, and resilience related?
Degeneracy	The ability of structurally different elements to perform the same function or yield the same output (Tononi et al., 1999). Two elements will seem functionally redundant, but when faced with disruptions, they seem functionally distinct. For example, in the human brain, many "neuronal groups can affect the output of any chosen subset of neurons in a similar way [...] a large number of different brain structures can influence, in series or in parallel, the same motor outputs, and after localized brain lesions, alternative pathways capable of generating functionally equivalent behaviors frequently emerge" (Tononi et al., 1999: 3257).	Supply chains that operate consistent with degenerate elements (i.e., many structures to one function) could be more resilient. Emphasis is on structurally different elements that allow trading organizations the flexibility to respond to disruptions. Degenerate elements may produce different outputs in different contexts (Edelman & Gally, 2001), such as trading organizations stepping up to do work in lieu of another. Degenerate supply chain structures could exhibit creativity, innovation, and evolvability (more than just stability against disruptions).	*Yes*—Degeneracy and potential organizational pluri-potentiality have not received much research attention in the context of organizational supply chains. *No*—research on related notions, including that companies' long-term success depends on current capabilities as much as exploring new competencies is not new (March, 1991). Work on ambidexterity and shared division of labor that renders supply chains more flexible has also received some research attention (e.g., Rojo et al., 2016).	*Yes*—Related research questions include: • How can the notion of degeneracy best be implemented, and when does it not make sense (at times, a focus on current capabilities and exploring new competencies is incompatible/not possible)? • What is the relationship between degeneracy and related notions, such as ambidexterity and shared division of labor? • What types of risks and disturbances (e.g., storms, droughts, global warming, and pandemics) are best handled through degeneracy?
Critical transitions	Critical transitions (also referred to as tipping points and thresholds) in natural systems are abrupt shifts in the integrity or state of a system caused by various environmental drivers	Disturbance can lead to critical transitions in supply chains too. Threshold-based concepts—adapted from natural sciences—can help us predict and react to significant changes in operational	*Yes*—The analogical equivalent of critical transitions in a supply chain context remains relatively unexplored. Predicting, avoiding, or recovering from transitions to undesired organizational	*Yes*—Related research questions include: • What's the nature of the interplay of external and internal factors driving critical supply chain transitions?

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	(Holling, 1973; Munson et al., 2018). The direction of these shifts is determined by an interplay of biotic and abiotic drivers (Scheffer et al., 2001). In natural lakes, for example, these transitions can result in different stable states—that is, clear or turbid lakes—depending on sediments, algal growth, runoff, shoreline erosion, and so on. Transitioning to deteriorated states can be catastrophic (Janssen et al., 2014).	environments (Allen et al., 2011). Research shows that the size of natural systems, spatial heterogeneity, and internal connectivity account for transitions to different states. Analogical parallels include the size and complexity of a supply chain and heterogeneity; these factors could be key in understanding, avoiding, or recovering from critical transitions to stable states following disturbances.	realities offers many novel research avenues. Much work on measuring and managing risk across the supply chain is based on discrete events rather than a gradual build-up of stress across system elements that can lead to sudden changes. *No*—research on critical transitions exists across fields beyond natural systems and includes global finance (think market crashes), medicine (think asthma attacks), and even SCM (think sustainable vs. unsustainable networks (Gruner & Power, 2020)).	• Can tipping points and their parameter(s) be identified and reversed (just like we have identified 100 degrees Celsius as the tipping point for water to change to vapor (see Gao et al., 2016))? • Can supply chain systems be organized to allow them to not only withstand disruptions but recover from transitions (or even reverse them so that systems are more robust than before)?

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Many scholars have turned toward natural phenomena (as have we in our illustration), yet, unlike natural phenomena, organizations are governed by people and social norms (see also Penrose's, 1952 critique of analogies in economics). Consider the example of Gruner and Power (2020), who turn to hydrology as a source domain to propose that supply chains' sustainability is binary: they are either sustainable or unsustainable. The authors argue that this is analogous to shallow lakes that are found to be either turbid or clear with no in-between states. But, when subjected to empirical inquiry, does this proposition hold, and if so, when? What boundary conditions might exist?

That said, no evaluative precaution can fully ensure that no prosaic ideas, dead ends, and even false theories are introduced when theorizing by analogy with creative intent. And yet, we have uncovered much support for the notion that it is worth the risk if we are to discover new insights that challenge what we know and help us tackle (emerging) SCM problems.

DISCUSSION AND IMPLICATIONS

Transferring insights from a target onto a source domain can lead to novel meaning structures via a juxtaposition (or "creative blend") of concepts from diverse fields (Cornelissen, 2005; Stephens et al., 2022; Turner & Fauconnier, 1999; Zahra & Newey, 2009). Transferring theoretical aspects in this manner can play a vital role in the early theorizing stage, where creative (and initially untested/unvalidated) insights are discovered. Still, this process often remains underused, messy, and somewhat mysterious, particularly in the discipline of SCM. Consequently, this article has integrated insights from research on analogies in areas such as organization theory, creativity, and design thinking to develop a heuristic-analytic model (e.g., Amabile, 1998; Auernhammer & Roth, 2021; Beckman & Barry, 2007).

Our model consists of clearly defined steps that can aid SCM scholars in deliberately generating creative theories. As a corollary, we help readers evaluate the rigor of proposed insights. If scholars follow and describe the proposed steps for reasoning by analogy, transparency in the often-mysterious early theorizing stage can be enhanced. When presented with more transparent accounts of the movement from source domain knowledge to creative theoretical insights in a target domain, readers can more easily evaluate the rigor of the final output (see Grodal et al. (2021) for a similar argument in the context of qualitative research).

Some organizational scholars argue against crossing disciplinary boundaries, often believing that it might slow down the field in the pursuit of its own paradigm (e.g., Markóczy & Deeds, 2009). In contrast, our heuristic-analytic model for the transfer of images between knowledge domains provides support and guidance for those intent on working at the intersection of disciplines to generate creative insights (e.g., Colquitt & Zapata-Phelan, 2007; Schmenner & Swink, 1998; Shaw et al., 2018; Zahra & Newey, 2009). As such, our work is consistent with the goal of stimulating fresh theoretical approaches to SCM, as evidenced, for example, by the *Journal of Supply Chain Management*'s Emerging Discourse Incubator series (2018-ongoing). Specifically, we enrich the small but growing body of work that promotes the adoption of analogical reasoning and creative knowledge transfer for the development of theoretical foundations in SCM (e.g., Chen et al., 2013; Foropon & McLachlin, 2013; Gruner & Power, 2020; Stephens et al., 2022).

Our model also challenges the often-purported notion that using analogies is overwhelmingly effortful. Instead, we propose that, when theorizing by analogy, cognitive processes can be both effortful/analytic and effortless/heuristic in equal parts (Kahneman, 2011). Indeed, viewing these two approaches as conflicting can overstate their differences while sacrificing significant complementarities. Without the analytic element, theorizing by analogy for creative insights can lead to little more than surface-level engagement with source domains, perhaps providing creative labels for phenomena but probably not adding meaningfully to SCM theory. In contrast, without the heuristic element, scholars can feel compelled to ignore stimulating domain differences, thus suppressing heterogeneity of thought and imaginative thinking under the guise of operating in a rigorous (or pseudo-rigorous), thorough, and more complete manner (Alvesson & Sandberg, 2020; Weick, 1989).

Finally, in focusing on supply chain resilience, we respond to calls for theory advances that can mitigate disruptive events such as those related to pandemics, natural disasters, wars, and so forth (e.g., Choi et al., 2020; Craighead et al., 2007, 2020; Govindan et al., 2017). Through our illustration, we hint at solutions to various supply chain disruptions. We do not propose that our generated insights are revolutionary. Instead, we intend to make a case for scholars to add analogical theorizing to their creative toolbox.

LIMITATIONS AND FUTURE RESEARCH

Our work has limitations that suggest fruitful research avenues. We largely neglect analogies' potential beyond

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their heuristic function in the early stage of theory development. For example, analogies can serve an explanatory or causal purpose in refining and qualifying theory (Cornelissen & Durand, 2014). Further research could explore different types of analogies and their role in developing theory (for an overview, see Ketokivi et al., 2017), while also examining their compatibility with the subjectivist–objectivist framework (Hassard, 1991). As noted, our model is primarily based on a subjective image transfer process. At the same time, analogies can prove helpful to more objectivist research approaches if the process can be used to develop specific propositions or hypotheses adequate for empirical testing and validation. In the grand scheme of things, our work makes only a modest contribution to the ongoing debate about how to make the most of analogies, serving to highlight significant potential for further applications.

Moreover, to minimize the introduction of trivial, prosaic, and false ideas, scholars should learn more about the role of evaluative criteria when theorizing by analogy. We hint at essential differences and critical evaluative criteria in the two key stages of theory development (see Figure 1). Still, more can be done to avoid inaccurate and misleading analogies (however creative) potentially propagating false insights and understandings. One criterion could be whether generated insights can be usefully transferred to other topics and adapted to a range of related circumstances (i.e., beyond a narrow target domain). More work is also needed on how other theorizing processes might help avoid analogies developing an ad hoc “cherry-picked” quality. Similarly, future research could shed light on the relationship between analogical theorizing and theory-developing qualitative research methods (grounded theory/abductive case methods, etc.), including when these processes act synergistically.

Also, in advocating a boundary-spanning knowledge search, we support studies that highlight the creative potential of domain dissimilarity in the analogical theorizing process (e.g., Oswick et al., 2002; Stephens et al., 2022). However, while we have addressed how and why domain dissimilarity can benefit scholars, much more nuanced insights are needed on when this is the case. For example, a more traditional comparison model of analogies based on similarity and analogous conditions (Chen et al., 2013; Tsoukas, 1991) could be more valuable than knowledge distance when analogies serve as a lens through which arguments (including empirical data) are examined beyond the early discovery stage of theorizing. Future research could measure and explore different levels of semantic distance between domains and how these levels might serve different purposes at different stages of the theorizing process.

Finally, in our illustration, interesting yet unanswered research questions emerged beyond those outlined in Table 1. How the insights generated with this process fare across different circumstances remains an open question. Our initial insights are unlikely to apply equally across different industries, for example. We can also not know whether other source domains and phenomena could have produced more creative insights. Even among the phenomena we excluded, promising insights might remain hidden. For example, the mentioned mathematical relationships and principles associated with harmony in music seemed more valuable in advancing knowledge on conflict resolutions among trading organizations than knowledge on resilience. Specifically, dissonance in music, an unpleasant combination of pitches, can balance out consonant sounds to create a pleasing musical effect (Chan et al., 2019). This begs the question of whether excluding this source domain could forfeit potential insights on the extent to which an interplay of tension and resolution (say, when it comes to critical resource allocations) are beneficial in fostering resilient supply chains.

Similarly, different source domains can yield contradictory findings. It remains unclear if these findings should be reconciled. Or does the playful, heuristic nature of the process make such steps unnecessary, as it mainly acts as inspiration for creative ideas?

In conclusion, we note that researchers have a remarkable capacity for creativity and combining knowledge domains seems particularly well suited to surfacing creative insights that advance scholarship and practice. This article’s objective is to inspire and guide this process to ultimately challenge or enhance what we know. While these advances on their own are likely sufficient motivation, we also found combining knowledge domains to be much fun!

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